



Tomato Leaf Disease Detection System

Computer Vision Based Disease Diagnosis

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Addressing Key Challenges in Agriculture

The Problem

- Manual inspection is slow and labor-intensive.
- Disease identification requires specialized expertise.
- Early detection is crucial to prevent widespread crop loss.
- Scalability issues for large-scale farming operations.

Our Impact

- Automated, rapid plant health monitoring.
- AI-driven diagnosis for 10 common tomato diseases.
- Proactive intervention to save crops.
- Efficiently monitors vast agricultural areas.



Project Overview: What Our System Does



Image Capture

Takes pictures of tomato leaves using cameras or uploaded images.



Leaf Detection

Accurately detects and isolates leaf regions within the image.



Disease Identification

Identifies the specific disease (if any) affecting the leaf.



Confidence Prediction

Provides a confidence score for each diagnosis, indicating reliability.



Real-time Monitoring

Enables continuous, automated monitoring of crop health.

Who Benefits from This System?



Farmers

Automate crop health monitoring and catch diseases early to save yields.



Agricultural Researchers

Study disease patterns and develop new solutions with accurate data.



Students

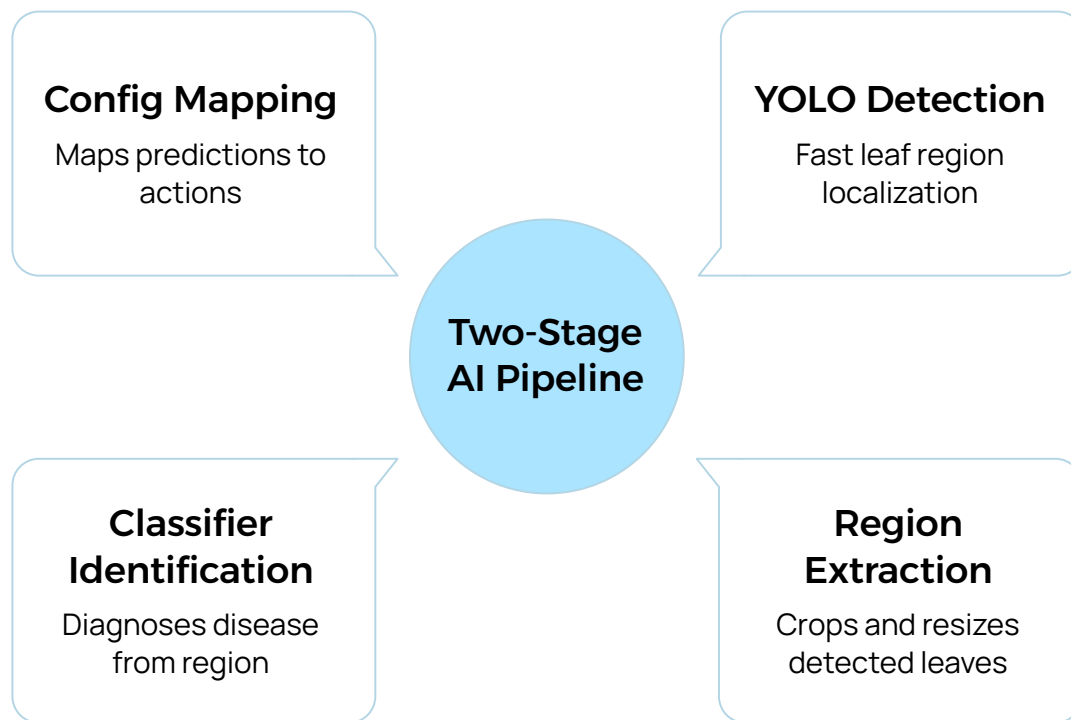
Gain practical experience in AI applications within the agricultural sector.



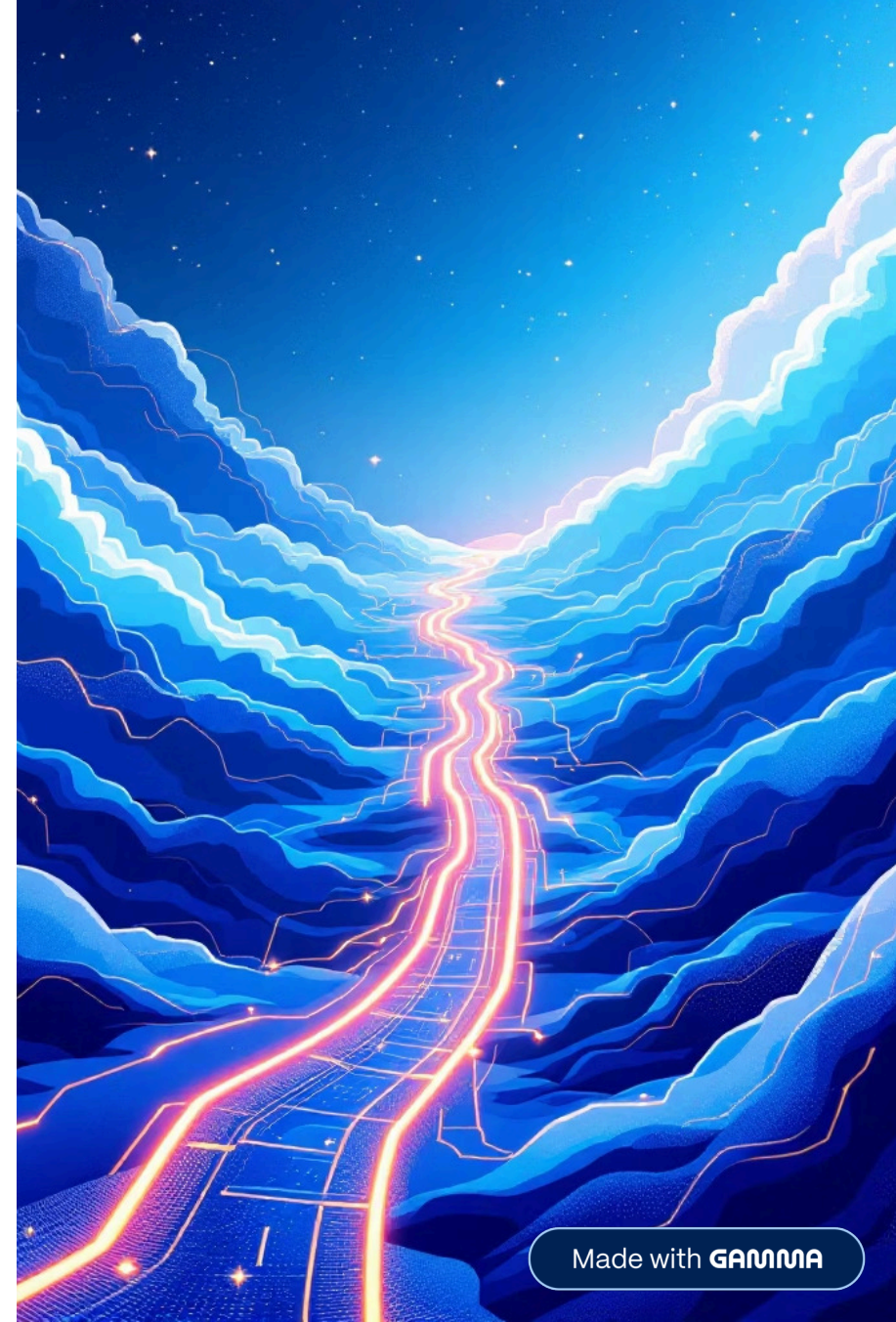
Developers

Deploy and integrate plant disease detection capabilities into new systems.

The Two-Stage AI Pipeline



Our system processes images in two distinct stages. First, YOLO quickly identifies and isolates leaf regions. Then, a sophisticated classifier analyzes these regions to determine the specific disease, using a configuration file to map numerical outputs to human-readable disease names.



AI Pipeline: Spotter and Doctor Analogy



YOLO: The Spotter

YOLO (You Only Look Once) acts as the "spotter," quickly finding and drawing bounding boxes around leaves in an image. It answers the question: "Where are the leaves?"



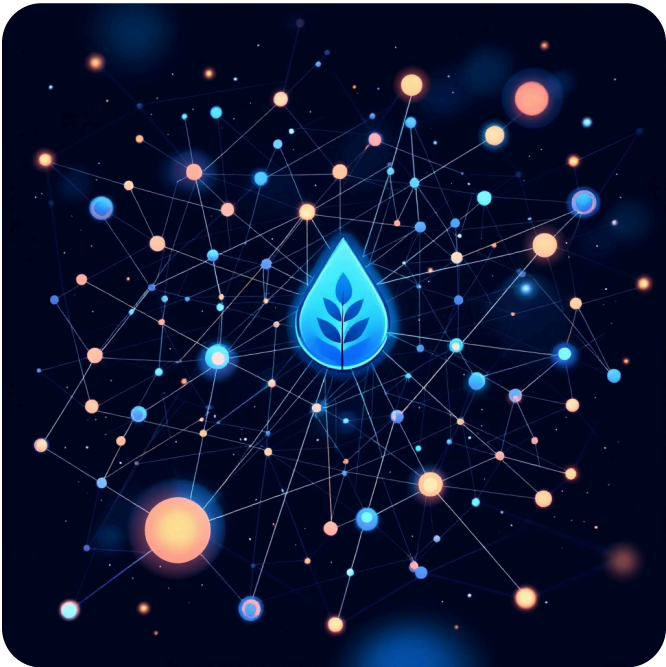
Classifier: The Doctor

The Swin Transformer model functions as the "doctor," taking the cropped leaf image and diagnosing the specific disease, providing a confidence percentage. It answers: "What disease does this leaf have?"

Core Components: The AI's Brains

A) YOLO Model (yolov8n.pt)

A pre-trained object detection model (~6 MB) that finds and draws boxes around plants/leaves. It's crucial for isolating the target before disease diagnosis.



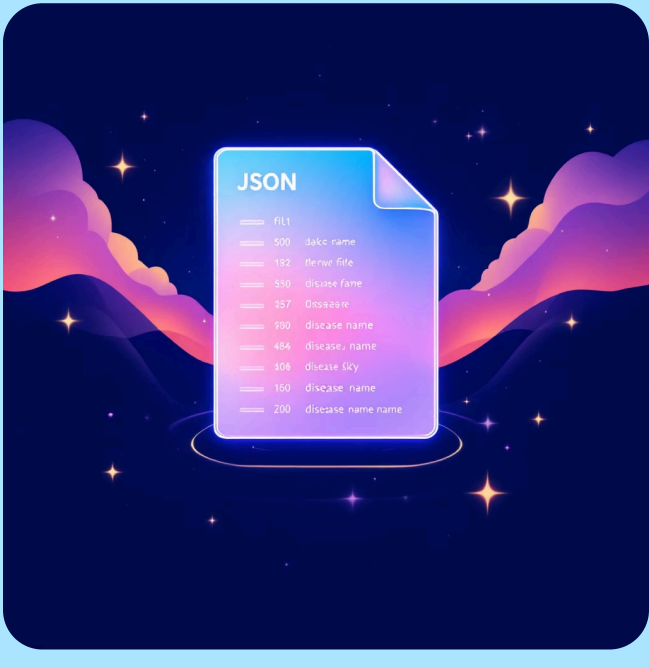
B) Disease Classifier (model.onnx)

A Swin Transformer model (~100 MB) trained on the PlantVillage dataset (11,000+ images) to predict 10 specific tomato diseases from cropped leaf images.



C) Configuration File (config.json)

A lookup table that maps the AI model's numerical outputs (0-9) to human-readable disease names, ensuring clear and understandable diagnoses.



YOLO Model Deep Dive: Leaf Detection



Key Features:

- **Model:** yolov8n.pt
- **Function:** Specializes in leaf detection.
- **Output:** Generates precise bounding boxes around identified leaves.
- **Performance:** Designed for fast inference, enabling real-time applications.
- **Purpose:** Essential first step to accurately isolate the area for disease classification.

Disease Classifier: Swin Transformer Power

Classifier Details:

- **Architecture:** Swin Transformer, optimized for image classification.
- **Training Data:** PlantVillage dataset, over 11,000 images of diseased and healthy tomato leaves.
- **Format:** ONNX model for efficient deployment and faster inference.
- **Workflow:** Takes cropped leaf images from YOLO and predicts one of 10 disease categories or "healthy."

Config File: The Translation Layer

The `config.json` file acts as a crucial lookup table within our system.

It specifically maps the numerical outputs generated by the AI model (from 0 to 9) to their corresponding human-readable plant disease names.

For example:

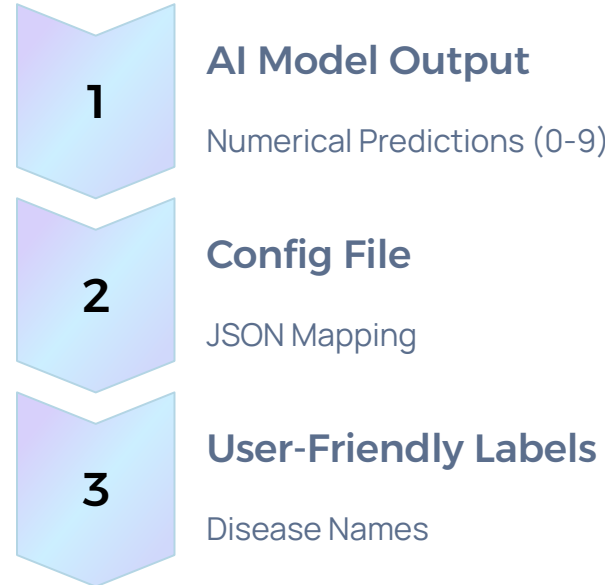
0 → Healthy

1 → Early Blight

2 → Late Blight

3 → Septoria Leaf Spot

This mapping serves as the critical translation layer, transforming raw AI predictions into clear, user-friendly diagnoses for effective interpretation.



Supported Diseases: 10 Tomato Leaf Conditions

Bacterial Spot

Early Blight

Late Blight

Leaf Mold

Septoria Leaf Spot

Spider Mites

Target Spot

**Tomato Yellow Leaf
Curl Virus**

Tomato Mosaic Virus

Healthy

Deployment Options: How to Use the System

Web App

Upload leaf images through a user-friendly web interface, accessible from any device.

- Accessible from any device
- No installation required
- Instant results

Live Camera Full

Real-time disease detection using a full camera feed for continuous monitoring.

- Continuous monitoring
- Process multiple leaves simultaneously
- Ideal for greenhouse automation

Live Camera Simple

Simplified camera interface for quick, single leaf analysis during field inspections.

- Easy to use
- Focused diagnosis
- Perfect for field inspections

Web Interface Workflow: Upload to Diagnosis



Upload

User uploads a leaf image through the web interface.



Preprocess

Image is prepared and normalized for AI analysis.



Predict

YOLO detection and Swin Transformer classification run.



Output

Results displayed with disease name and confidence score.

Live Camera Full Pipeline: Real-Time Monitoring



Frame Capture

Camera continuously captures video frames from the greenhouse or field.



YOLO Detection

Detects all leaves in the frame, drawing bounding boxes around them.



Crop Regions

Extracts individual leaf regions from the detected bounding boxes.



Classify

Swin Transformer analyzes each cropped leaf for disease identification.



Draw Results

Displays bounding boxes with disease labels and confidence on the live feed.

Simplified Camera Mode: Single Leaf Analysis

Whole Frame Mode



Analyzes the entire camera frame at once for plant health. This mode provides a quick overview, ideal for initial assessments of multiple leaves in a field or greenhouse for rapid scanning and monitoring.

Color-Based Selection



Allows the user to select a specific leaf by color or region of interest. It's designed to focus on a particular plant or area, enabling targeted diagnosis of specific plants with precision.

Grid-Based Selection



Divides the camera frame into a grid for systematic analysis. This mode facilitates methodical inspection of organized crop rows, making it best for structured greenhouse or field layouts.

Training Pipeline: From Data to Deployment



Dataset

PlantVillage dataset with 11,000+ labeled tomato leaf images



Augmentation

Data augmentation techniques applied to increase dataset diversity and robustness



Training

Model training using the augmented dataset with optimization and validation



Export ONNX

Trained model exported to ONNX format for efficient deployment

CNN Training Overview: Building the Classifier

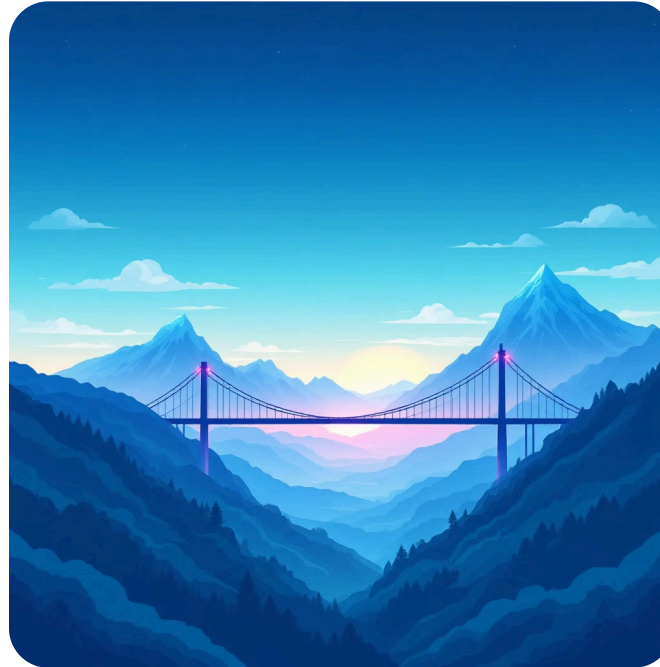
SwinV2 Architecture

- **Key Feature:** Hierarchical attention mechanism.
- **Advantage:** Enhanced performance on image classification.



Transfer Learning

- **Key Feature:** Utilizes ImageNet pre-trained weights.
- **Advantage:** Faster convergence and better generalization.



Data Augmentation

- **Key Feature:** Rotation, flipping, brightness/color adjustments.
- **Advantage:** Improves model robustness and prevents overfitting.



YOLO Training Workflow: Object Detection Model

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1. Dataset Labeling

Annotate tomato leaf images with bounding boxes.

Details: Each leaf region is manually labeled with precise coordinates.

Tools: YOLO-compatible annotation format (YOLO txt format with normalized coordinates).

Dataset: Thousands of labeled leaf images from various angles and lighting conditions.

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2. Training Steps

Train the YOLO model on the labeled dataset.

Details: Model learns to detect leaf regions through iterative optimization.

Process: Forward pass → Loss calculation → Backpropagation → Weight updates.

Hyperparameters: Learning rate, batch size, epochs, augmentation parameters.

Validation: Regular validation on held-out test set to monitor performance.

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3. Output Weights

Save trained model weights as `yolov8n.pt`.

Details: Lightweight model (~6 MB) optimized for real-time inference.

Format: PyTorch weights file ready for deployment.

Performance: Fast inference speed suitable for edge devices and live camera feeds.

Complete Data Flow: End-to-End Pipeline

This comprehensive workflow illustrates how raw input from users or sensors is transformed through various AI models to deliver actionable insights on plant health.



Challenges & Troubleshooting: Common Issues & Solutions

Deploying AI models in real-world agricultural settings presents unique challenges. Addressing these issues systematically ensures robust and reliable performance from detection to classification.

YOLO Detection Issues

- Missed Leaf Detection
- False Positives
- Overlapping Leaves

Lighting & Environmental Problems

- Poor Lighting Conditions
- Leaf Reflections & Glare
- Dust & Debris on Lens
- Motion Blur

Performance Bottlenecks

- Slow Inference Speed
- High Memory Usage
- Network Latency (Cloud Deployment)
- Model Accuracy Degradation



Conclusion: Transforming Agriculture with AI

Our journey into AI-powered agricultural solutions culminates in a system designed to revolutionize farming practices, making them more efficient, sustainable, and productive. This presentation outlined our current achievements and a clear roadmap for future innovation.

Key Takeaways

AI-Powered Disease Detection

- Automated diagnosis of 10 tomato leaf diseases with high accuracy.
- Real-time monitoring capabilities reduce manual inspection time by 90%.
- Enables early intervention, significantly preventing crop loss.

Two-Stage Pipeline Architecture

- YOLO for precise leaf localization, ensuring focal accuracy.
- Swin Transformer for robust and accurate disease classification.
- Optimized for both inference speed and diagnostic accuracy.
- Deployable across various platforms, from web to edge devices.

Flexible Deployment Options

- Intuitive web interface for quick analysis and reporting.
- Live camera systems for continuous, automated monitoring.
- Edge deployment for reliable offline operation in remote areas.
- Scalable from single farms to extensive agricultural operations.